[INCH-POUND]
A-A-50682
June 23, 1989

COMMERCIAL ITEM DESCRIPTION

SCALAR NETWORK ANALYZER
10 MHz TO 40 GHz FREQUENCY RANGE

The General Services Administration has authorized the use of this Commercial Item Description by all federal agencies.

Scope. This Commercial Item Description (CID) describes a Scalar Network Analyzer. This CID is meant as a minimum requirement for the Scalar Network Analyzer in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

Applicable Documents.

Specifications, Standards, and Handbooks. The following specifications, standards, and handbooks form a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-T-28800 Test equipment for use with electrical and electronic equipment, general specification for.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Engineering Division, SA-ALC/TILDD Kelly AFB TX 78241-5000.

MISC N/A FSC 6625

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Other Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/IEEE C62.41

ANSI/IEEE 488 - 1978

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Order of Precedence. In the event of conflict between the text of this specification and the references cited herein (except for the associated detail specifications, specification sheets for MS standards) the text of this CID take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Salient Characteristics. The equipment shall be capable of operation within the accuracy's, limits, and specifications herein.

1. Safety. The equipment shall be comply with the type III safety requirements of MIL-T-28800.

a. Leakage Current Protection. Protection shall be provided from leakage current in excess 3.5 milliamperes (mA) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(s).

b. Radio Frequency Interference (RFI) Capacitors. Any capacitor used in RF1 filters shall be rated for at least four (4) times the operating voltage.

c. Type III Ventilation Openings.

(1) Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts.

(2) Top ventilation openings shall incorporate small mesh screening or similar design to prevent the explosion of hot electronic particles due to failure of electronic devices such as capacitors.

2. Parts, materials, and processes. Except for marking, the SNMS shall comply with the type III parts, materials, and process requirements of MIL-T-28800. Front panel marking shall be in accordance with MIL-T-28800.

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a. Restricted materials. The equipment shall comply with the restricted material requirements of MIL-T-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3. Design and construction. Unless otherwise specified herein, the equipment shall comply with the type III design and construction requirements of MIL-T-28800. The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits where construction is susceptible to leakage or corona, indicators and associated controls shall be in close proximity to each other, all potentiometer controls shall have at least a 270 degree rotation range (4.71s radians).

a. Type of Construction. Construction of the SNMS shall be of a monolithic or a separate component design.

b. Modular Design Restrictions. The modular design concept for the SNMS is acceptable with the following restrictions:

(1) Each module shall contain a power circuit protection fuse.

(2) Each module plug-in card shall be keyed and mated with the back plane receptacle so that modules may not be randomly placed into the mainframe.

(3) Modules shall not be removable by access through the front or rear of the mainframe with the unaided hand.

(4) A package part number shall be assigned to the SNMS as a whole system. This system part number shall be marked on the mainframe identification plate and all plug-in(s).

(5) All module interfaces shall be done in the back of the mainframe.

c. Signal Input and Output Connectors. Unless otherwise specified, all signal input and output connectors shall be type BNC female.

d. Internal Cooling. The equipment shall be capable of operating continuously for twenty-four (24) hours at 55'C. If forced air is used, it shall be as specified in MIL-T-28800.

(1) Forced Air Cooling Motor. DC brushless motors shall be utilized for the forced air cooling motor in the equipment.

(2) Air Filter. Equipment utilizing a cooling scheme which exhausts air from the equipment through the cooling fan shall not require an externally accessible air filter.

(3) Thermal Switch. A thermal switch shall be provided which will interrupt circuit power to the Scalar Network Analyzer when the internal temperature rises to the degree of possible damage to

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the electronic components. A minimum of 10°C hysteresis drop from this tripping temperature must occur before circuit power is again applied.

4. Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 110, 120, 200, 220 and 240 volts ac (+10%) at line frequencies of 50 Hz to 400 Hz.

a. Electrical Power Connectors. The equipment shall operate with a ground at the primary source on either side of the single phase input.

b. Power Consumption. Maximum power consumption of all components combined shall not exceed 450 VA.

c. Fuses. Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements.

d. Surge and Transient Voltage Protection. The SNMS shall contain surge and transient voltage protection internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per type III solid state tolerance band of MIL-T-28800. The surge voltage protection circuit shall conform to ANSI/IEEE Std. C62.41-1980. The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection. Normal operation of the Scalar Network Analyzer may be inhibited and false operational signals may be generated under input voltage transients greater than +20 percent and frequency transient conditions greater than +5 percent-from within the nominal type III steady-state tolerance bands of MIL-T-28800. Damage (other than fuse failure) shall not occur during voltage transient conditions greater than +20 percent and frequency transient conditions greater than +5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) specified in MIL-T-28800. Normal operation of the Scalar Network Analyzer shall be automatically resumed when the transient has passed and any blown fuses are replaced. Blown fuses shall be externally accessible. Testing shall be in accordance with MIL-T-28800.

5. Dimensions and Weight. The size and weight of individual component shall be consistent with current commercial capabilities, refer to paragraphs 5.a and 5.b of this CID.

a. Equipment Dimensions. The nominal dimensions per component for the SNMS should be approximately 200 mm height x 490 mm wide x 500 mm deep.

b. Equipment Weight. The weight including all accessories (excluding manual) per component, shall not exceed 50 pounds.

6. Enclosure Requirements. Unless otherwise specified herein, the equipment shall comply with the style E enclosure requirements of MIL-T-28800.

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a. Rack mounting. Rack mounting capability shall be available as an option for the equipment specified herein.

b. Tilt-Bail Handle. A tilt-bail handle shall be provided in lieu of the hinged metal handles in accordance with MIL-T-28800. Exception to this paragraph may be acceptable as this is not a prime consideration of the equipment.

c. Reusable Pouch or Container. An attached accessory pouch may be used for storage of all supplied accessories in lieu of front-cover storage.

7. Marking and Identification.

a. Marking. The visible surface adjacent to panel facilities such as variable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation hereof, indicating the purpose or use of the part.

(1) Panel Marking. The front panel marking shall comply with the type III parts, materials, and processes panel marking requirements of MIL-T-28800; however, marking identification processes other than the type III requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with this CID.

b. Items to be Marked (if applicable).

(1) Volume and Weight (including accessories) shall be marked on three external surfaces of the enclosure.

(2) The type designation of the AN nomenclature of the SNMS shall be stenciled on the exterior or main frame on at least three adjacent surfaces, on the transient case on two opposite sides. Marking shall be in block letters approximately 2 inches in height.

8. Environmental Requirements.

a. Temperature Requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class 5 temperature requirements specified by MIL-T-28800 and as listed below.

(1) Non-Operating Temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -40°C to +650°C.

(2) Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of O°C to 50°C.

b. Humidity Requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for class equipment as specified by MIL-T-28800.

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c. Altitude Requirements. The equipment shall meet its required performance and accuracy requirements under the altitude requirements for class 5 equipment as specified by MIL-T-28800.

d. Vibration Requirements. The equipment shall meet the class 5 Vibration Requirements as specified by MIL-T-28800.

e. Mechanical Shock Requirements.

(1) Shock, Functional. The equipment shall meet the class 5 shock, pulse requirement as specified by MIL-T-28800.

(2) Transit Drop. The equipment shall meet transit drop requirement as specified by MIL-T-28800.

(3) Bench Handling. The equipment shall meet the requirements as specified by MIL-T-28800.

(4) Shock, High Impact. The equipment shall meet the requirements as specified by **M**IL-T-28800.

f. Electromagnetic Interference. The equipment shall comply with MIL-T-28800, Table XII.

g. Magnetic Environment. The equipment shall comply with the magnetic environment requirements of MIL-T-28800.

9. Reliability. Reliability verification testing, shall be in accordance with the type III, class 5 environmental conditions of MIL-T-28800.

a. Mean Time Between Failures (MTBF). The equipment Mean Time Between Failures (MTBF) shall be 2,000 hours.

b. Pre-Conditioning Requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

(1) Burn-In. Burn-in shall expose the equipment to extended periods of elevated temperatures, (40'C for a minimum of 100 hours). Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of MTBF. Prior to and following burn-in, a Satisfactory Operation Test (SOT) shall be performed. A Satisfactory Operation Check (SOC) shall be performed conducted at selected times during the burn-in test.

(2) Thermal Cycling. Thermal Cycling shall expose the equipment to hot and cold cycle for Class 5 equipment per MIL-T-28800. Each equipment which fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the cycling, a

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Satisfactory Operation Check (SOC) shall be conducted during the last two upper and lower temperature limits.

10. Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case of modules.

a. Maintainability. The equipment shall meet the type III maintainability requirements specified by MIL-T-28800.

11. Interchangeability. The equipment shall meet the type III interchangeability requirements specified by MIL-T-28800.

12. Digital Interface. The equipment shall meet the digital interface requirements per MIL-T-28800.

13. Performance Characteristics. Unless otherwise specified, all vernier controls shall be in the calibrated position and all performance requirements shall apply with or without the probes, after a 30 minutes warm-up period in a laboratory ambient operating temperature.

a. Display Presentation. The equipment shall provide a display presentation with a minimum 8 divisions by 10 division display.

(1) Graticule. The display shall have the capability to display nonparallax graticule markings. Subdivision markings of .2 divisions shall be provided along the vertical and horizontal center axis. The 10 and 90 percent points of a 5 centimeter or greater vertical display shall etched across the horizontal axis is readily discernable dots or marks. This applies to an etched graticule only.

b. CRT Controls/Adjustments. Unless provided through some internal or preset, means, the following CRT controls and adjustments shall be provided.

(1) Intensity. A front panel control capable of furnishing optimum intensity of the displayed signal shall be provided. The intensity control shall provide a continuous range of control from an extinguished beam to at least greater than optimum intensity.

c. CRT/Front Panel Active Readouts. The CRT shall display but not be limited to the following active readouts. The Scaler Network Analyzer shall have a 2 channel display. The active readout capabilities apply to both channels.

Reference level
Frequency
Scale/div
Power

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(1) Reference Level Control. The reference level shall be controlled from the front panel and displayed on the CRT.

d. Active Function Controls. The scalar network analyzer shall have at least the following active function controls located on the front panel.

Offset controls
Averaging and Normalizing controls
Reference Line controls
Marker controls

(1) Measurements. The scalar network analyzer shall have the capability of absolute power and ratio measurements over 10 megahertz to 26.5 gigahertz frequency rating.

e. Impedance. The input impedance shall be 50 ohms nominal.

f. Resolution. The Scalar Network Analyzer shall have independent resolution in steps of 10, 5, 1 and .5 and .2 or .1 db/div. independently for each channel.

g. Signal Averaging. Averaging algorithms shall be implemented to improve the input signal-to-noise ratio. The number of averages shall be selectable in a binary sequence from 4 to 256 (i.e. 2, 4, 8, 16, 32, 64, 128, 256).

h. Memory. The scalar network analyzer shall have non-volatile memory capable of storing up to 9 complete test set-ups for over 30 days after the last turn-off of the system.

i. Testing and Diagnostics. The equipment shall provide the capability perform automatic self-tests and extended diagnostic routines to diagnose the majority of internal electrical failures. In addition, the equipment shall provide internal self-calibration and extended calibration routines which are executable from the front panel and the FPIB interface. If required signals may be manually connected to the signal inputs.

j. General-Purpose Interface Bus. The equipment shall be fully programmable and controllable via the IEEE Standard 488-1978 General-Purpose Interface Bus (GPIB). A GPIB compatible connector shall be provided on the equipment rear panel. With the exception of power on/off, all equipment controls and equipment functions, including diagnostic and calibration capabilities shall be controllable via the GPIB interface.

k. Adaptive Sweep. The scalar network analyzer shall automatically set sweep speed (i.e. fast sweep speed at high power levels and a slower sweep speed at low levels).

1. Sweep Input. The scalar network analyzer shall have a sweep input range of at least 0.0 volts to 10.0 volts nominal.

m. Functions. The equipment shall have, as a minimum, the following functions:

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(1) Setting, storage, and recall of front panel settings.

(2) Storage and recall of reference waveforms.

(3) Marker control/positioning.

(4) Internal diagnostic and calibration routines.

n. Waveform Transfer. The equipment shall provide the capability to transmit and receive digitized waveforms via IEEE-488-1978 interface. Waveforms sent to the equipment shall be treated as reference waveforms and shall be loaded into the appropriate reference waveform memory. In addition, waveform scaling data shall also be transferable over the GPIB interface and displayable on the CRT with the reference waveform(s).

o. Plotter Output. The equipment shall provide the capability to directly drive an external X-Y or IEEE-488-1978 plotter without the requirement for an instrument controller. This capability shall allow the generation of hard copy from displayed waveforms.

p. Printer Output. The equipment shall provide the capability to directly drive an external or an IEEE-488-1978 printer or X-Y Plotter without the requirement for an instrument controller. This capability shall allow the generation of hard copy from displayed waveforms.

q. Required Accessories. The following (minimum) accessories shall be required with each deliverable equipment. The accessories' shall be compatible with the vendor's perspective equipment.

Operator's Manual(s)
Spare fuse(s)
US Power Cord (at least 2.0m, 6 ft.)
Three (3) RF Diode Detectors
Direction Bridge
Two (2) Waveguide Adapters (Size [EIA] WR 112)
Two (2) Waveguide Adapters (Size [EIA] WR 90)
Extension Cables

(1) RF Diode Detector. The RF Diode Detector shall meet the following minimum requirements.

A. Frequency Range: 0.01 to 26.5 Gigahertz

B. Maximum Power Input at least +20 dBm (lOOmW)

C. Input Impedance shall be 50 ohms nominal

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(2) Extension Cables. Extension Cables shall be available to extend the distance between the scalar network analyzer and the detector or bridge to a maximum of 200 ft. without degradation of performance.

(3) Directional Bridge. A directional bridge shall be provided which meets the following minimum requirements.

A. Frequency Range: 0.01 to 26.5 Gigahertz

B. Directivity: At least 35 dB

C. Input Impedance: 50 ohms

D. Test Port Connector: Type N

(4) Waveguide Adapters: Two sizes of waveguide adapters shall be provided which meet the following minimum requirements.

A. Waveguide Size WR 112:

(1) Frequency Range: 7.05 GHz - 10.0 GHz

(2) SWR: Not more than 1.25

B. Waveguide Size WR 90:

(1) Frequency Range: 8.20 GHz - 12.4 GHz

(2) SWR: Not more than 1.25

14. Calibration Interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

Packaging, packing, and marking. The items shall be packed in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with national motor freight classification and uniform freight classification. Preservation, packaging, packing, labeling, and marked shall be a specified in the contract or order.

Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, and standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer

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products with established commercial marketplace acceptability, may compete by submission of Bid Samples. (The Government reserves the right to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.)

Known Acceptable Products. The following product(s) are known to have previously met the requirements of this CID and need not to resubmit samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Manufacturer	Model	NSN
Hewlett-Packard	8756A	6625-01-176-8505
Wavetek	1038-Nl0	6625-01-144-0852

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MILITARY INTEREST:

Custodian
Air Force - 99
Navy – SH

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS

Preparing Activity
Air Force - 82

DoD Project 6625-F891

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